

DEREK GREENE

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Shady Cove, OR | Open to Remote Positions

SUMMARY

Software Engineer with strong backend in full-stack development, currently pursuing an Honors B.S./M.S. in Computer Science with a 4.0 GPA. Experienced in Python, Flask, Elixir, and web security research. Built and deployed several web applications and contributed to research in domain control validation. Open to remote software engineering roles to apply and expand my technical expertise.

EDUCATION

Oregon State University — M.S. Computer Science *(Expected Apr. 2027)*
GPA: 4.0 | Alpha Sigma Lambda Honor Society

Oregon State University — Honors B.S. Computer Science *(Expected Apr. 2026)*
GPA: 3.97 | Honors Thesis | Alpha Sigma Lambda Honor Society

Rogue Community College — A.S. Computer Science *(Apr. 2024)*
GPA: 3.93 | Member, Honor Society

PROJECTS

TOTP Authentication System | *Elixir, Phoenix*

- Developed a Time-based One-Time Password (TOTP) authentication system using Elixir and Phoenix framework.
- Implemented secure token generation and validation with proper cryptographic standards.
- Created a user-friendly interface for account setup and authentication management.

Stake Website | *Python, Flask, HTML/CSS*

- Designed and deployed a dynamic website for church operations, integrating bot detection and secure user access.
- Maintained infrastructure and supported stakeholder updates.

CT Domain Data & DomainScout | *Python, Flask, Research*

- Built a tool to analyze domain control validation vulnerabilities using Certificate Transparency logs.
- Created a GUI to surface vulnerable domains with disposable WHOIS contact methods.
- Authored a research paper for submission to the Internet Measurement Conference.

Engineering Services Database | *Flask, MySQL, Jinja2*

- Developed a full-stack CRUD web application to manage service records using Python, Flask, and MySQL.
- Implemented intuitive user interface and secure database operations.

Low-Level IO in Assembly | *x86 Assembly*

- Created an interactive CLI tool to parse, validate, and compute operations on signed decimal inputs.

- Handled input validation, conversion, and basic arithmetic with user prompts.

TECHNICAL SKILLS

Languages: PHP, Python, Elixir, C, C++, Bash, x86 Assembly, SQL, HTML/CSS, JavaScript

Tools & Frameworks: Flask, Phoenix, MySQL, SQLite, MongoDB, React, Git, Docker, AWS, GCP, Node.js, Express, Mix, Jupyter, CMake, Postman

Platforms: Linux, Windows, macOS

Other: Windows System Administration, Networking Troubleshooting

EXPERIENCE

Rentec Direct — Software Developer

Oct 2025 - Present

- Troubleshooting a pre-existing applications driven by PHP/MySQL
- Design and develop new features
- Code and database maintenance
- Peer review & testing of pre-release code

Oregon State University — Cybersecurity Research Fellow

Apr 2024 - Present

- Conduct research in cybersecurity with focus on domain control validation and web security vulnerabilities.
- Collaborate with faculty on academic research projects and paper publications.
- Contribute to ongoing security research initiatives within the computer science department.

COMMUNITY SERVICE

- Volunteer, ARUP Utah Domestic Violence Coalition 5K
- Co-Chair, International Humanitarian Law, Red Cross Volunteer Services
- Volunteer, Primary Children's Hospital & Bingham Creek Library
- Member, ARUP Lab Week Planning Committee